

20592 - Probability and Statistics

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Final project

Deadline: January 10

1 The datasets

In this project, we consider two different datasets. A short description of the datasets is given in the following.

CreditCard dataset. This dataset consists in a cross-section data on the credit history for a sample of applicants for a type of credit card. This is a relatively large dataset, in particular, you have

- A binary variable **card**, which gives if the application for a credit card was accepted or not.
- Other 9 covariates of interest.
- 2 covariates not to consider: **share** and **expendure**; you can discuss in the report why these two covariates may be discarded.

This dataset is available directly in the library **AER** of R.

azcabgptca dataset. This dataset consists in a subset of the 1991 Arizona Medicare data for patients hospitalized subsequent to undergoing a cardiovascular procedure. This dataset is composed in particular of:

- A count variable **los**, which is the number of days stayed in the hospital.
- Other 5 covariates of interest giving information about the patient and the procedure.

This dataset is available directly in the library **COUNT** of R.

You can also find all the data in the folder dedicated to the final project. A more elaborate description of the two datasets is given in the txt file named “Datasets explanation”. There are two goals, one for each dataset. For the **CreditCard** dataset, it is to assess the impact of the covariates on the access to a credit card. This will be studied using a logistic regression. For the **azcabgptca** dataset, it is to assess the impact of the covariates on the number of doctor visit, which should be done using a Poisson regression.

2 Assignment

Submission. The project is assigned and submitted in groups. The submission consists of two items:

- A .pdf file, the report of the project. It can NOT contain any code. The length of the report shall be around 3-4 pages.
- A Rmarkdown or Jupyter notebook, containing all the codes that reproduce the results in your report.

Each of the datasets should be studied separately using the corresponding regression. For each of these problems:

1. (a) Create a code script implementing the Fisher scoring algorithm (use the canonical link function for Poisson regression).
(b) Fisher scoring and Newton's method coincide for canonical link functions. Prove this (in the report). Hint: derive the Hessian of the log-posterior from scratch and observe the definitions of Fisher scoring and Newton's method. (The last question below is about non-canonical links.)
2. Create a code script which implements a Markov Chain Monte Carlo method (e.g. random walk Metropolis-Hastings) for respectively a Bayesian logistic regression and a Bayesian Poisson regression.
3. Apply all your scripts to the datasets and discuss the results.

The report should include:

- A description of the algorithms, with their relevance for GLMs.
- The analytical derivation of all the quantities used in the algorithms (updating equations, acceptance probabilities, etc.)
- Basic diagnostics for the convergence of the MCMC, you notably should provide a trace-plot and autocorrelation plot for the parameter associated with respectively `income` and `procedure`.
- A discussion of the prior distributions chosen in point 2.
- A final discussion on the fitted models (interpretation of the parameters, etc.), in particular regarding the role of the covariates. Point out the three most influential covariates for the outcome and interpret the results with words.
- A discussion on the following question: here we only consider GLMs (logistic and Poisson regression) with a canonical link, what will change in the Fisher scoring algorithm if a non-canonical link was considered? A well-known example of such a GLM is called the *probit regression* which is an alternative to logistic regression for binary data.

Remark 1. The algorithms should be implemented from scratch (without using for example the function `FisherScoring` in R, but you can use non-specialized functions like `diag` for example). You can use either R or Python.

Remark 2. The scripts should run on our computers, without any additional information. Failure to reproduce the results on the report will lead to zero points awarded.

Remark 3. You should understand the ideas required to be able to implement this assignment from scratch. The final written exam will test you on this understanding. Any signs that the code or report is AI generated will be viewed harshly when graded.